

A gap between recorded time and usable knowledge

Research-gap document · worked author example · 2 October 2026

The question

On a fixed relational autocomplete task, how much does measured information availability change the apparent value of relational context? The question is narrower than “Do relational models work?” It asks whether a specific comparison survives when a predictor can use only information actually available at its query cutoff.

The course mission is to test whether relational learning unlocks value that strong single-table pipelines miss. That argument requires a credible information contract. A sophisticated model cannot repair a comparison whose two arms see different information.

Why this is a candidate gap

RelBench v2 introduces temporally constrained autocomplete. The inherited L181 report records a training-health blocker for its selected GNN lane. L184 records a source/temporal gate. These are local evidence boundaries, not proof that the research community has neglected availability.

Our proposed contribution would be a **paired availability audit and controlled comparison**, conditional on obtaining arrival histories and a healthy predictor. The contribution is not a new architecture by declaration. If related work already supplies the same contract and experiment, the novelty claim must be narrowed or abandoned.

What this document establishes

The accompanying L190 replay authenticates saved artifacts, reconstructs a complete selected model comparison, and reproduces incomplete literature coverage. It also makes an authored priority rubric inspectable. It does not run a new model or establish novelty.

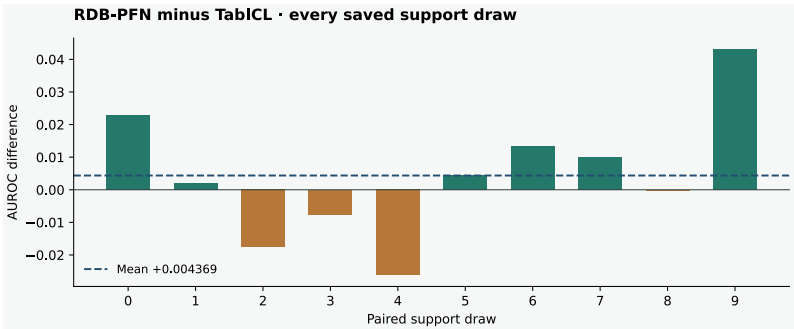
Decision: investigate availability first, because the question is precise and the next missing input is identifiable. Do not launch the proposed experiment until its data, implementation and cost gates pass.

What the evidence actually says

Saved model	Mean AUROC	Sample SD across support draws
RDBPFN	0.721938	0.019983
RDBPFN_single	0.663990	0.034177
TabICLv1.1	0.717568	0.009770

The named comparison is RDB-PFN v5 Table 9, F1/driver-dnf: three released models, ten paired support draws, 512 support rows and all 702 test queries. The L190 replay authenticates and re-scores **21,060 saved predictions**. It verifies query/label identity, complete run coverage, paired support sets and support/test separation against released prepared arrays.

The RDB-PFN minus TabICL paired mean is **0.004369 AUROC**, positive in **6 of 10** draws. The plot shows every paired difference. These draws reuse one task and test population; their dispersion is not independent cross-database uncertainty.



Measured AUROC differences on the same frozen test population. All ten paired support draws shown; six are positive. No database-level confidence claim.

The comparison supports a narrow descriptive sentence about these saved predictions. It does not test a composite hybrid. The RDB-PFN source uses a relational synthetic prior and a DFS interface; RelGNN already supplies composite message passing. An interaction between the two remains a proposal.

Literature coverage: raw L188 responses yield **31 records, 30 unique papers, and 2 of 4 completed searches**. Four failed attempts remain in the packet. Successful replay preserves the incomplete search; it does not turn missing results into evidence of absence.

Boundaries: released labels/DFS are trusted inputs here; raw label reconstruction, historical arrival histories, new inference and pretraining are not performed. Other lesson reports provide attributed status receipts only; their numerical experiments are not replayed in L190.

Three questions worth separating

Candidate	Impact	Data / implementation / compute	Priority
Availability-aware autocomplete	4	5 / 4 / 5	18.667
Composite structure in a foundation learner	5	3 / 2 / 4	15.000
Accuracy under serving and privacy constraints	4	3 / 3 / 2	10.667

Rubric: impact $\times$ weighted mean of data, implementation and compute feasibility. Each authored score is ordinal, from 1 to 5; larger feasibility means easier. Default weights are equal. Across all 27 triples from $\{1,2,3\}$, availability remains first. Weight stability does not validate the assigned scores or predict scientific success.

This checkpoint substitutes joint serving/privacy constraints for L189's pretraining-transfer candidate. Its shortlist and composite contrast are separate authored choices, not a replay of L189's ranking.

1 · Availability-aware autocomplete

Hypothesis: relational-context benefit shrinks under measured arrival-time visibility. Compare the same predictor and queries under an event-time policy and an availability policy. Existing keys and audits lower implementation effort; absent arrival histories can still block the whole study. A null or reversed effect would weaken the proposed failure mode.

2 · Composite structure in a foundation learner

Hypothesis: composite routes interact positively with relational pretraining. Use a matched four-arm design: conventional/composite encoder $\times$ scratch/pretrained initialization. Compare the interaction, not only whether the combined arm wins. Architecture changes may invalidate checkpoint reuse; no compatible pretrained hybrid is established by our saved scores.

3 · Accuracy under serving and privacy constraints

Hypothesis: a relational pipeline retains useful accuracy while meeting fixed freshness, latency and entity-level privacy requirements. Compare it with a strong flattened baseline under the same workload and accounting. L186's course simulation and L187's privacy experiment motivate requirements, but do not establish joint real-world performance. Any mandatory constraint failure defeats the joint claim.

Choice: availability first; composite structure second; joint constraints third. All three retain NOT_ESTABLISHED novelty and aggregate execution cost. These rankings are an authored decision aid, not permission to train.

An experiment that could change our mind

Proposed paired study · not executed

Task: the L181 F1 results-position autocomplete contract. Preserve the released train/validation/test split, complete query population, target masking and proxy exclusions. Pin the task, database, source and preprocessing hashes before any fitting. Require recorded availability timestamps for every consumed feature; missing histories block the historical study.

Arms: a strong flattened baseline and a relational predictor, each under (A) event-time visibility and (B) event-time plus measured availability visibility. Fit each policy using its own permitted inputs. Keep architecture, preprocessing rules, tuning allowance and validation selection paired across A/B; missing context follows one declared rule. Repair and re-audit the inherited training-health issue first. Exact compatible model configuration remains an admission deliverable, not an implied fact.

Replicates: prospective seeds 0, 1 and 2 for every arm. Select checkpoints on validation only; evaluate every held-out query once after freezing choices. Retain per-query predictions, full keys, visibility masks, per-seed checkpoints, failed attempts and all preparation/training/validation costs.

Estimand and decision

Let benefit under policy P be $\text{MAE}(\text{flat}, P) - \text{MAE}(\text{relational}, P)$. The primary contrast is $\Delta = \text{benefit}(A) - \text{benefit}(B)$. Positive Δ means the apparent relational advantage shrank when measured availability was enforced. Pair queries and training seeds throughout.

Before execution, adopt **0.10 finishing-position units** as an illustrative course threshold for a useful shrinkage, and justify or revise it with the intended stakeholder. Use 1,000 paired driver-cluster bootstrap resamples, RNG seed 19001, retaining all cutoffs for each sampled driver. Report per-seed results separately; the interval describes sampled-driver variation on this task, not database transfer.

Interpretation rule: lower 95% interval bound above 0.10 supports the declared useful shrinkage; upper bound at or below 0.10 fails that criterion; otherwise the result is inconclusive. A negative Δ challenges the direction. Do not replace missing histories with fabricated dates. Synthetic delays may support a separately labelled mechanism experiment.

Limits, gates and the next decision

Four limits that remain visible

Scientific coverage: the quarterly search is incomplete. Inspect disconfirming prior work and its methods before claiming novelty. A local source bug or absent artifact does not prove an open scientific problem.

Historical validity: a released event timestamp is not an ingestion log. The current replay does not reconstruct raw labels or DFS, and it cannot prove what a historical predictor knew. Arrival evidence is the first admission gate for the proposed study.

External validity: the saved comparison uses one fixed task and population. Support draws measure sensitivity to supplied examples. They are neither new databases nor independent benchmark replications.

Execution readiness: exact compatible model configuration, healthy training and an aggregate cost forecast remain unresolved. A high ranking cannot waive them. Serving/privacy simulations do not certify a live joint utility claim.

Budget and stop rules

Executed L190 lane: saved-evidence replay and validation only; \$0 new cloud/API spend; 1,800 aggregate local execution seconds including failed attempts, notebook execution and checks. Stop and preserve **INCOMPLETE** if the cap is reached. The mutable attempt ledger is separate from the immutable evidence packet.

Future study: standing aggregate cap about \$10, including preparation, all arms/seeds, selection, evaluation, retries and verification. Cost is currently **NOT_ESTABLISHED**; this document authorizes no paid run. Do not reduce the design after seeing costs or results while retaining the original claim.

Checkpoint verdict and learner task

The selected evidence replay is complete. The research checkpoint remains **INCOMPLETE**: novelty, experiment admission and the learner's written defense are separate gates.

Submit your own five-page argument. Replace the author judgments with justified choices; identify the strongest existing counterexample; explain what result would reverse your ranking. The teacher must review the reasoning. A notebook pass cannot establish research judgment or personal mastery.

Next cheap decision: determine whether genuine arrival histories can be obtained under the task contract. If not, retain the historical study as blocked and explicitly propose a narrower mechanism study for separate approval.